Notes: Chetty, Dobbie, Goldman, Porter & Yang (QJE, 2026)

Changing Opportunity: Sociological Mechanisms Underlying Growing Class Gaps and Shrinking Race Gaps in Economic Mobility

Contents

1 Big picture

- **Question.** How and why did intergenerational mobility change across recent U.S. birth cohorts by race and parental-income class?
- **Core fact.** Between the 1978 and 1992 cohorts, the white class gap widened while the low-income white–Black race gap narrowed.
- **Main data contribution.** The paper links federal tax records, census records, Numident mortality data, ACS outcomes, and SAT/ACT scores for roughly 57 million children born between 1978 and 1992.
- **Primary outcome.** Children's household income rank in adulthood, measured relative to individuals in the same birth cohort and calendar year.
- **Mechanism.** Changes in childhood environments, proxied by race–class–county parental employment rates, explain the divergent trends.
- **Causal design.** Movers across counties identify childhood exposure effects because younger movers are exposed longer to destination environments.
- **Social interaction.** Outcomes are most related to parental employment rates of peers children are likely to interact with: own cohort, own group, and high-interaction cross-race groups.
- **Takeaway.** Economic mobility can change rapidly within a generation; social environments transmit parental-generation shocks to children.

2 Data and measurement

2.1 Linked administrative data

The paper combines federal income tax returns, decennial census records, Numident mortality data, ACS variables, and linked SAT/ACT records. The sample covers approximately 57 million children born between 1978 and 1992.

Interpretation: The data allow the authors to measure outcomes, parents, and childhood places for a very large set of cohorts and subgroups.

2.2 Race, class, and cohort definitions

Class is defined by parental income during childhood. The key class points are the 25th and 75th percentiles of the national parental income distribution. The main race comparison is between white and Black children because these groups generate the central widening class gap and shrinking low-income race gap.

Interpretation: Defining class by parental income is essential because divergent trends remain even conditional on parental education.

2.3 Outcomes

The main outcome is household income rank in adulthood. The paper also examines employment, mortality, education, SAT/ACT participation, and SAT/ACT scores among test takers.

Interpretation: Similar patterns across monetary and nonmonetary outcomes imply that the changes are not purely a measurement artifact of adult household income.

2.4 Community measures

The key environmental measure is parental employment rates in communities defined by county, race, class, and cohort. In the baseline, parental employment is measured when children are age 27, using W-2 records, but robustness checks use other measures and timing choices.

Interpretation: Parental employment is interpreted as a proxy for broader childhood social and economic environments, not only the employment of a child's own parents.

2.5 Mover and sibling samples

The causal design uses children who move exactly once across counties during childhood. The within-family analysis compares siblings who move together but differ in age and therefore exposure duration.

Interpretation: Movers and siblings are central because they help separate causal exposure from cross-sectional place differences and family selection.

3 Models and empirical specifications

3.1 Class and race gaps

$$ClassGap_s = \bar{y}_{s,White,p=75} - \bar{y}_{s,White,p=25}, \qquad RaceGap_s = \bar{y}_{s,White,p=25} - \bar{y}_{s,Black,p=25}.$$

Interpretation: The paper studies how these gaps change between the 1978 and 1992 birth cohorts.

3.2 Family and neighborhood controls

$$y_i = \alpha + \beta_1 HighIncome_i + \beta_2 \frac{s_i - 1978}{14} + \beta_3 HighIncome_i \frac{s_i - 1978}{14} + Controls_i + \varepsilon_i.$$
$$y_i = \alpha + \beta_1 White_i + \beta_2 \frac{s_i - 1978}{14} + \beta_3 White_i \frac{s_i - 1978}{14} + Controls_i + \varepsilon_i.$$

Interpretation: The interaction coefficient measures the change in the class or race gap after holding fixed observed family or neighborhood factors.

3.3 Community parental employment relationship

$$\Delta \bar{y}_{cpr} = \alpha + \beta \Delta \bar{e}_{cpr} + \varepsilon_{cpr}.$$

Interpretation: This is the descriptive mechanism regression relating changes in child outcomes to changes in parental employment rates in the same community.

3.4 Mover exposure effect

$$\bar{y}_{dprsm} = \eta_{dprs} + (A - m)\mu_{dprs} + \theta_{dprsm}.$$
$$\Delta \bar{y}_{dpr,m=0} - \Delta \bar{y}_{dpr,m=A} = A \Delta \mu_{dpr}.$$

Interpretation: Comparing younger and older movers across cohorts identifies the causal exposure effect if changes in selection do not vary with move age.

3.5 Social interaction regressions

$$\bar{y}_{cprs} = \alpha_{cpr} + \alpha_{spr} + \sum_k \beta_k \bar{e}_{cpr,s+k} + \varepsilon_{cprs}.$$

$$\Delta\bar{y}_{cpr} = \beta_w \Delta\bar{e}_{c,p=25,White} + \beta_b \Delta\bar{e}_{c,p=25,Black} + \varepsilon_{cpr}.$$

Interpretation: These specifications test whether outcomes respond most to the employment rates of socially proximate peers.

4 Identification and empirical design

4.1 Descriptive trends

The first layer documents national trends by race and class. This establishes the puzzle but is not causal.

Interpretation: The descriptive facts motivate the mechanism analysis.

4.2 Controls for family and neighborhood factors

The paper then asks whether observed family characteristics or stable neighborhoods explain the trends. They do not: family controls explain only a small share, and neighborhood controls do not remove the central patterns.

Interpretation: This motivates the search for changing community-level environments.

4.3 Community parental employment

Changes in children's outcomes are strongly correlated with changes in parental employment rates in the same race–class–county community. Adding these rates to the gap regressions explains the white class gap increase and much of the race gap decline.

Interpretation: This is the main mechanism correlation but does not alone prove causality.

4.4 Mover exposure design

The paper's causal design uses children who move across counties. Younger movers receive more exposure to the destination environment than older movers. Comparing the cohort-change gradient by move age identifies the causal effect of changing childhood environments.

Interpretation: This design uses duration of exposure as the source of identification.

4.5 Selection tests

The paper tests the constant-selection-by-age assumption using own-parent employment restrictions, origin–destination fixed effects, adult-county fixed effects, parental-income-change controls, and predicted outcomes based on pre-move family characteristics.

Interpretation: Similar estimates across these checks support, but do not mechanically prove, the causal interpretation.

4.6 Within-family design

Sibling comparisons hold fixed many family factors. If younger siblings benefit more from moving to improved environments than older siblings, this supports childhood exposure effects.

Interpretation: Sibling evidence is especially helpful because it directly addresses family-level selection.

4.7 Social interaction mechanism

The own-cohort, own-group, exposure, and interracial-marriage tests show that parental employment matters most for groups children are likely to interact with.

Interpretation: This supports a social-interaction channel rather than only broad county-level resources.

5 Tables and figures

5.1 Figure I: Intergenerational Mobility by Birth Cohort, Race, and Class

Read: The figure shows widening white class gaps and shrinking low-income white–Black race gaps across cohorts.

Detailed interpretation: Low-income white children fall relative to high-income white children, while Black children from low-income families improve relative to low-income white children.

Economic message: This is the headline fact the rest of the paper explains.

5.2 Table I: Mean Child Household Income Rank by Birth Cohort, Race, and Class

Read: The table compares child income ranks at parental income percentiles 25 and 75 by race/ethnicity.

Detailed interpretation: The strongest changes occur for white and Black children, motivating the paper's focus on these groups.

Economic message: The table converts the figure into transparent numbers and shows which groups drive the main facts.

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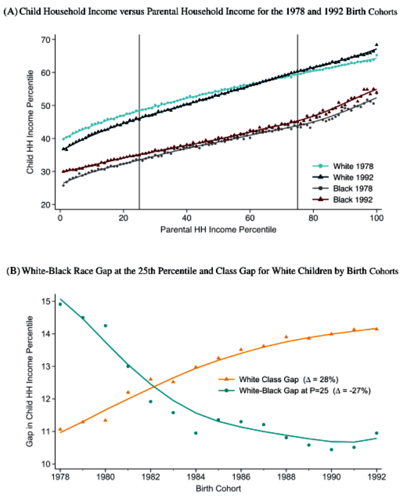

FIGURE I

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TABLE I

MEAN CHILD HOUSEHOLD INCOME RANK BY BIRTH COHORT, RACE, AND CLASS

	Mean HH income rank at P = 25				Mean HH income rank at P = 75			
	1978 cohort (1)	1992 cohort (2)	Change (3)	Share (%) (4)	1978 cohort (5)	1992 cohort (6)	Change (7)	Share (%) (8)
White children	48.4	46.1	– 2.3	22.8	59.5	60.2	0.8	38.3
Black children	33.5	35.1	1.6	10.3	43.9	45.3	1.4	3.2
Hispanic children	44.3	44.7	0.4	9.9	53.1	53.6	0.4	4.3
Asian children	51.6	51.6	0.0	1.6	57.4	58.1	0.7	1.7
AIAN children	35.2	35.7	0.5	0.6	47.0	51.3	4.3	0.3

Notes. This table reports the change in mean household income rank for children in the 1978 and 1992 birth cohorts. Columns (1) and (2) report mean household income ranks for children born to families at the 25th percentile of the national income distribution in the 1978 and 1992 cohort, respectively; column (3) reports the change in mean household income rank for these children; column (4) reports the share of all children who are in the indicated race group and born to families with below-median incomes; columns (5) and (6) report mean household income ranks for children born to families at the 75th percentile of the national income distribution in the 1978 and 1992 cohort, respectively; column (7) reports the change in mean household income rank for these children; and column (8) reports the share of all children who are in the indicated race group and born to families with above-median incomes. We estimate mean household income ranks using fitted values from a lowess regression on parental income percentiles for each race and birth cohort. See Figure I for additional details

5.3 Figure II: White Class and White-Black Race Gaps in Pre-Labor Market and Nonmonetary Outcomes by Birth Cohort

Read: The figure studies employment, mortality, education, SAT/ACT participation, and test scores.

Detailed interpretation: Patterns appear before full adult labor-market entry, so the mechanism is not just adult earnings measurement.

Economic message: The evidence broadens the paper from income mobility to opportunity more generally.

5.4 Figure III: The Changing Geography of Intergenerational Mobility for Low-Income Families

Read: The maps show county-level changes in mobility for low-income white and Black children.

Detailed interpretation: Geographic mobility patterns are persistent but not fixed; changes vary across places and groups.

Economic message: This motivates the county-level community environment analysis.

5.5 Figure IV: White Class and White-Black Race Gaps Controlling for Family- and Neighborhood-Level Factors

Read: Controls for parental education, wealth, occupation, marital status, county, and tract effects do not eliminate the trends.

Detailed interpretation: Observable family changes explain only a small share of the widening white class gap.

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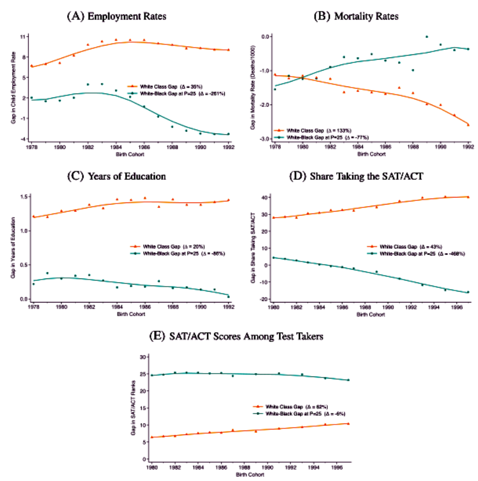

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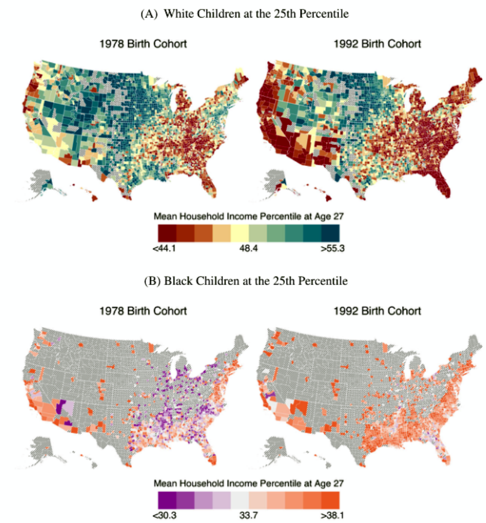

FIGURE III

The Changing Geography of Intergenerational Mobility for Low-Income Families

Economic message: The figure redirects attention toward changing community-level factors rather than static neighborhood or family characteristics.

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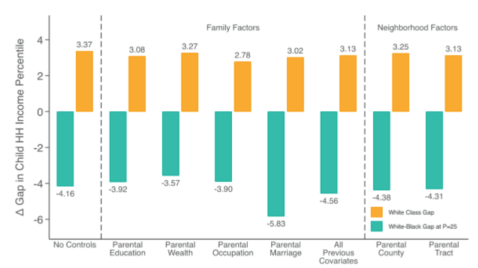

FIGURE IV

White Class and White-Black Race Gaps Controlling for Family- and Neighborhood-Level Factors

This figure reports reports OLS regression estimates of the change in the white class and white-Black race gaps, controlling for family- and neighborhood-level factors. The first pair of bars reports the change in the white class and white-Black race gaps between the 1978 and 1992 birth cohorts with no controls, estimated by regressing children's household income ranks on a linear cohort control interacted with class (for the white class gap) or race (for the white-Black race gap). The next five pairs report estimates controlling for family-level factors interacted with class and cohort fixed effects (for the white class gap) or race and cohort fixed effects (for the white-Black race gap). The seventh pair reports estimates controlling for county fixed effects interacted with class and cohort (for the white class gap) or race and cohort (for the white-Black race gap). The final pair reports estimates controlling for census tract fixed effects interacted with class and cohort (for the white class gap) or race and cohort (for the white-Black race gap). For the white class gap, we restrict the sample to white children born to fam-

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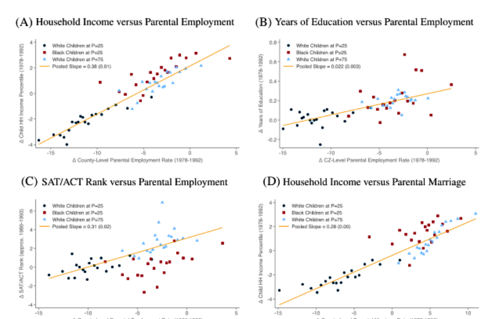

FIGURE V

Changes in Children's Outcomes in Adulthood Versus Changes in Parental Outcomes by Race, Class, and County

These figures show binned scatterplots of changes in children's outcomes in adulthood versus changes in parental outcomes for white and Black children born to families at the 25th percentile of the national income distribution and white children born to families at the 75th percentile of the national income distribution.

5.6 Figure V: Changes in Children's Outcomes Versus Changes in Parental Outcomes by Race, Class, and County

Read: The figure links changes in child outcomes to changes in parental employment and marriage rates.

Detailed interpretation: The income, education, and test-score panels show strong positive slopes with parental employment.

Economic message: This is the central mechanism correlation connecting community parental employment to child outcomes.

5.7 Figure VI: Identifying the Causal Exposure Effect of Changes in Childhood Environments

Read: The figure explains the mover identification design.

Detailed interpretation: Younger movers provide more childhood exposure to the destination; older movers help net out destination selection and adult labor-market conditions.

Economic message: This is the conceptual causal design behind the paper.

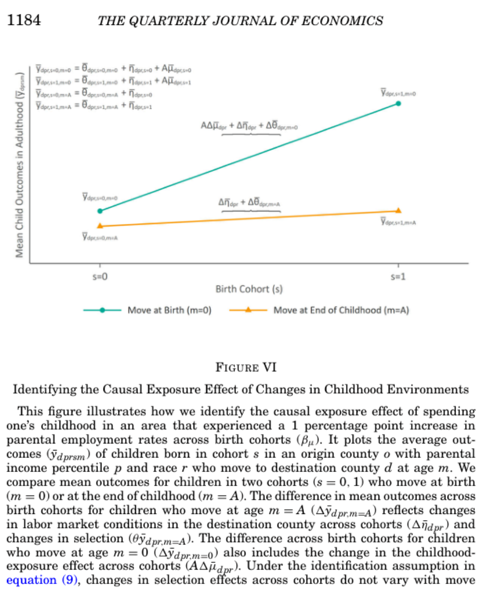
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FIGURE VI

Identifying the Causal Exposure Effect of Changes in Childhood Environments

This figure illustrates how we identify the causal exposure effect of spending one's childhood in an area that experienced a 1 percentage point increase in parental employment rates across birth cohorts (β_μ). It plots the average outcomes ($\bar{y}_{dprsm}$) of children born in cohort s in an origin county o with parental income percentile p and race r who move to destination county d at age m. We compare mean outcomes for children in two cohorts ($s = 0, 1$) who move at birth ($m = 0$) or at the end of childhood ($m = A$). The difference in mean outcomes across birth cohorts for children who move at age $m = A$ ($\Delta\bar{y}_{dpr,m=A}$) reflects changes in labor market conditions in the destination county across cohorts ($\Delta\bar{\eta}_{dpr}$) and changes in selection ($\theta\bar{y}_{dpr,m=A}$). The difference across birth cohorts for children who move at age $m = 0$ ($\Delta\bar{y}_{dpr,m=0}$) also includes the change in the childhood-exposure effect across cohorts ($A\Delta\bar{\mu}_{dpr}$). Under the identification assumption in equation (9), changes in selection effects across cohorts do not vary with move

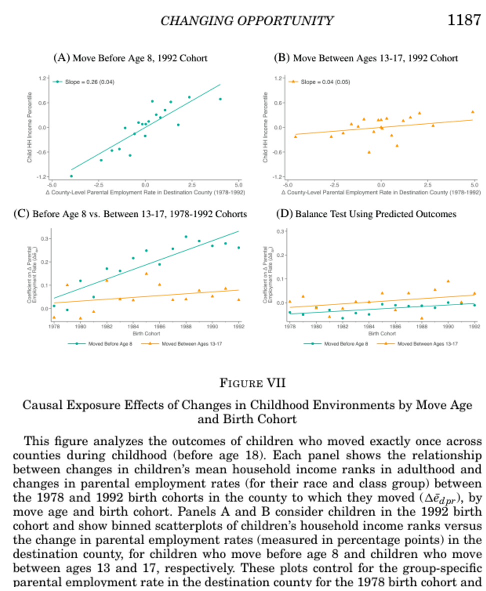
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FIGURE VII

Causal Exposure Effects of Changes in Childhood Environments by Move Age and Birth Cohort

This figure analyzes the outcomes of children who moved exactly once across counties during childhood (before age 18). Each panel shows the relationship between changes in children's mean household income ranks in adulthood and changes in parental employment rates (for their race and class group) between the 1978 and 1992 birth cohorts in the county to which they moved ($\Delta\bar{e}_{dpr}$), by move age and birth cohort. Panels A and B consider children in the 1992 birth cohort and show binned scatterplots of children's household income ranks versus the change in parental employment rates (measured in percentage points) in the destination county, for children who move before age 8 and children who move between ages 13 and 17, respectively. These plots control for the group-specific parental employment rate in the destination county for the 1978 birth cohort and

5.8 Figure VII: Causal Exposure Effects of Changes in Childhood Environments by Move Age and Birth Cohort

Read: Children moving earlier to improving destinations have better outcomes than those moving later.

Detailed interpretation: The relationship weakens with move age, consistent with exposure duration.

Economic message: This is the main mover evidence for causal childhood exposure effects.

5.9 Table II: Childhood-Exposure Effect of Changes in Parental Employment Rates: Estimates Based on Movers

Read: The table reports baseline mover estimates and robustness checks.

Detailed interpretation: The estimates remain positive after controls designed to address own-parent employment, adult-location effects, and selection on observables.

Economic message: The table is the main quantitative causal evidence.

TABLE II

CHILDHOOD-EXPOSURE EFFECT OF CHANGES IN PARENTAL EMPLOYMENT RATES: ESTIMATES BASED ON MOVERS

	Child household income rank					Predicted rank	
	Semi-parametric estimates (1)	Origin × destination FE (2)	Own par. employed (3)	Adult county FE (4)	Δ Par. income (5)	Using early par. income (6)	Adding par. chars. (7)
Δ Par. emp. dest. × scaled cohort × scaled move age (β_μ)	0.339				0.338	0.039	0.031
	(0.097)				(0.106)	(0.021)	(0.063)
Δ Par. emp. dest. (γ_0)	−0.016				0.004	0.026	0.009
	(0.031)				(0.035)	(0.008)	(0.030)
Δ Par. emp. dest. × scaled cohort (γ_1)	0.059				0.037	0.023	0.019
	(0.070)				(0.081)	(0.012)	(0.040)
Δ Par. emp. dest. × scaled move age (γ_2)	0.064				0.015	−0.092	−0.076
	(0.058)				(0.055)	(0.015)	(0.049)
Exposure to parental employment (β_μ)		0.273	0.280	0.445			
		(0.080)	(0.098)	(0.052)			
Origin × par. inc. × race × cohort × move age FE	X			X	X	X	X
Dest. 1978 par. emp. × cohort × move age FE	X				X	X	X
Origin × dest. × par. inc. × race × move age FE		X	X				
Par. inc. × race × cohort FE		X	X				
Par. inc. × race × cohort × adult county FE				X			
Change in parent income around move					X		
Number of children (1,000s)	5,213	3,090	1,449	3,973	4,407	5,213	857

TABLE II

CONTINUED

Notes. This table reports OLS regression estimates of β_μ, the causal effect of growing up from birth in a community with a 1 percentage point higher parental employment rate on children's household income ranks in adulthood. The estimation sample includes children who moved across counties once during childhood. We limit the sample to children with origin and destination counties with more than 2,000 children in the same race and class (defined as above- or below-median parental household income) group, between the 1978 and 1992 cohorts. Column (1) reports estimates of β_μ from equation (10). We regress children's household income ranks in adulthood on the change in race-by–parental income percentile–specific parental employment rates between the 1978 and 1992 birth cohorts in the destination county ($\Delta\bar{e}_{dpr}$). To construct the semi-parametric analogue of Figure VII, we interact $\Delta\bar{e}_{dpr}$ with move age, cohort, and move age times cohort. We rescale move age and cohort as in equation (10), so that they each range from zero to one in our sample. We also control for origin county-by-race-by–parental income percentile–by–birth cohort–by–move age fixed effects and cohort-by–move age fixed effects interacted with the group-specific parental employment rate in the destination county for the 1978 birth cohort. In columns (2)–(4), we regress children's household income ranks on mean race-by–parental income percentile–by–cohort-specific parental employment rates across origin and destination counties, weighted by the number of years the child spent in each county during childhood. Column (2) reports estimates of β_μ from Online Appendix equation (C.1). We control for race-by–parental income percentile–by–origin county–by–destination county–by–move age fixed effects and race-by–parental income percentile–by–birth cohort fixed effects. Column (3) repeats this specification but restricts to children whose parents are employed when the child is age 27. Column (4) controls for origin county-by–parental income percentile–by-race-by–birth cohort–by–move age fixed effects and adulthood county–by–parental income percentile–by-race-by–birth cohort fixed effects. Adulthood county is where the child is living at age 27. Column (5) repeats the specification in column (1), but adds controls for deciles of the change in parent income rank from two years before to two years after the move. Column (6) repeats the specification in column (1) using the predicted child household income rank based on parent income rank early in childhood (from birth to age four) as the outcome. Column (7) repeats column (6), adding the following pre-move parental characteristics to the prediction model for children's income ranks: education, occupation, wealth, and marital status. See Section V.C for additional details on how we construct these predicted income ranks. All specifications calculate county-by-subgroup parental employment rates using non-movers and parents who move more than once following the procedure described in Online Appendix A. Standard errors, clustered by origin county, are reported in parentheses. See Section II for details on the variable definitions and Section V for details on the sample construction. All statistics are cleared under Census DRB release authorizations CBDRB-FY2023-CES005-025, CBDRB–FY24–0359, and CBDRB–FY25–028.

5.10 Figure VIII: Causal Exposure Effects of Changes in Childhood Environments: Within-Family Estimates

Read: Sibling comparisons use younger-minus-older sibling outcome differences.

Detailed interpretation: The effect is strongest when siblings are farther apart in age, exactly when exposure differences are larger.

Economic message: This supports the mover design by holding many family characteristics fixed.

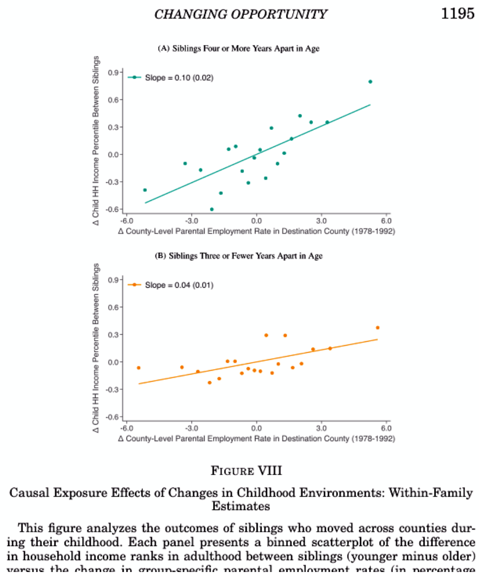

FIGURE VIII

Causal Exposure Effects of Changes in Childhood Environments: Within-Family Estimates

This figure analyzes the outcomes of siblings who moved across counties during their childhood. Each panel presents a binned scatterplot of the difference in household income ranks in adulthood between siblings (younger minus older)

FIGURE IX

Children's Outcomes in Adulthood Versus Parental Employment Rates by Relative Birth Cohort

This figure shows how children's outcomes in adulthood correlate with parental employment rates in their own versus adjacent birth cohorts. The green series in circles reports estimates from an OLS regression of children's household income ranks in adulthood on group-specific parental employment rates in one's own birth cohort and adjacent birth cohorts, controlling for county-by-race-by–parental income percentile fixed effects and cohort-by-race-by–parental income percentile fixed effects, as in equation (11). We weight by the number of children in each county-by-race-by-class cell. We restrict the sample to low-income white and Black children and high-income white children in the 1982–1988 birth cohorts and counties with more than 2,000 children in the relevant race and parental income group (split by above- or below-median household income) across the 1978 and 1992 birth cohorts. The vertical bars denote 95% confidence intervals, with standard errors clustered at the county level. See Online Appendix A for details

5.11 Figure IX: Children's Outcomes in Adulthood Versus Parental Employment Rates by Relative Birth Cohort

Read: Own-cohort parental employment is most predictive of adult outcomes.

Detailed interpretation: Adjacent cohorts matter less, suggesting the relevant peer environment is cohort-specific.

Economic message: This supports social interaction rather than only county-wide resource channels.

5.12 Figure X: Changes in Children's Outcomes Versus Changes in Parental Employment Rates by Degree of Social Interaction

Read: Outcomes are most related to employment rates of groups children are likely to interact with.

Detailed interpretation: White parental employment matters more for Black children when exposure to white peers or interracial interaction is higher.

Economic message: The figure provides the strongest evidence for a social-interaction mechanism.

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FIGURE X

Changes in Children's Outcomes in Adulthood Versus Changes in Parental Employment Rates by Degree of Social Interaction

This figure reports OLS regression estimates of the effect of county-level changes in parental employment rates on county-level changes in children's household income ranks in adulthood. Panel A shows estimates of the effect of changes in low-income white (green bars) and low-income Black (orange bars) parental employment rates on changes in outcomes for low-income white children and low-income Black children. We restrict the sample to white and Black children born to families at the 25th percentile of the national income distribution. Panels B and C report estimates of the effect of changes in low-income white (green bars) and low-income Black (orange bars) parental employment rates on changes in outcomes for low-income Black children. Here we restrict the sample to Black children born to families at the 25th percentile of the national income distribution. Panel B examines heterogeneity by Black children's exposure to white people in

Figure X: Degree of Social Interaction. The figure shows that social proximity predicts whose parental employment rates matter for children's outcomes.

6 Short summary

- Class gaps among white children widened, while low-income white–Black race gaps narrowed.
- The patterns appear in income, education, test-score, employment, and mortality outcomes.
- Standard family and static neighborhood controls explain little of the trends.
- Race–class–county parental employment rates explain most of the divergent changes.
- Mover and sibling designs support causal childhood exposure effects.
- Social proximity evidence points to interaction and social capital as key mechanisms.

7 Expanded conclusion

7.1 Core conclusion

Economic opportunity changed sharply within one generation. The widening white class gap and narrowing low-income white–Black race gap show that mobility is not fixed solely by deep historical forces.

7.2 Mechanism

The main mechanism is changing childhood social environments, proxied by parental employment rates in race–class–county communities. Mover evidence indicates that exposure to improving communities causally improves outcomes.

7.3 Social interaction

The social-interaction evidence is central: children respond most to peers and parents with whom they are more likely to interact. This shifts the interpretation from places alone to social communities within places.

7.4 Policy interpretation

The paper points toward policies that support youth in communities experiencing parental-generation shocks and that foster cross-class and cross-race interaction, mentorship, social capital, and opportunity networks.

7.5 Limitations

The paper identifies broad environment effects, not a single intervention. Parental employment is a proxy for a bundle of social, economic, institutional, and cultural changes. The mover design is powerful but still depends on assumptions about selection by move age.

7.6 Takeaway

The paper's enduring message is that economic mobility can change rapidly when childhood social environments change. To improve mobility at scale, policy must address the social structure of opportunity, not only household income or neighborhood geography.